

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

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Experiment No.: 5-1

1. TITLE

Interactive Product Calculator

2. AIM

To create an interactive webpage containing a product name, quantity, and price, and dynamically calculate the total price whenever the quantity or price changes without refreshing the page.

3. OBJECTIVE

The objectives of this experiment are:

- To understand JavaScript functions and variables.
 - To use DOM manipulation to access and update HTML elements.
 - To use event listeners to respond to changes in user input.
 - To calculate the total price dynamically.
 - To validate quantity and price values.
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4. THEORY

JavaScript is a scripting language used to make web pages interactive by responding to user actions and changing webpage content dynamically.

The Document Object Model (DOM) represents an HTML document as objects that JavaScript can access and modify. In this application, JavaScript obtains the quantity and price values from the input elements, multiplies them, and displays the result in the total-price element.

Event listeners detect changes in the quantity and price fields, so the calculation is updated immediately without reloading the webpage.

Formula: Total Price = Quantity × Price per Unit

5. ALGORITHM / PROCEDURE

1. Create an HTML webpage with fields for product name, quantity, and price.
2. Create an element to display the total price.
3. Define a JavaScript function named calculateTotal().
4. Read the quantity and price values from the DOM.
5. Validate the quantity and price values.

6. Multiply quantity by price.
 7. Display the calculated total in the webpage.
 8. Attach input event listeners to the quantity and price fields.
 9. Call calculateTotal() when the page loads.
 10. Open the webpage in a browser and test different inputs.
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6. PROGRAM

index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Interactive Product Calculator</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

  <div class="calculator">

    <h1>Product Calculator</h1>

    <div class="form-group">

      <label for="productName">Product Name:</label>

      <input type="text" id="productName"

        placeholder="Enter product name">

    </div>

    <div class="form-group">

      <label for="quantity">Quantity:</label>

      <input type="number" id="quantity"
```

style.css

```
* {  
  box-sizing: border-box;  
}  
  
body {  
  margin: 0;  
  min-height: 100vh;  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  font-family: Arial, sans-serif;  
  background-color: #f2f2f2;  
}  
  
.calculator {  
  width: 400px;  
  padding: 30px;  
  background-color: white;  
  border: 1px solid #ccc;  
  border-radius: 8px;  
}  
  
h1 {  
  margin-top: 0;  
  text-align: center;  
}  
  
.form-group {  
  margin-bottom: 18px;  
}  
  
label {
```

6. PROGRAM (CONTINUED)

script.js

```
function calculateTotal() {  
  
    const quantity = Number(  
  
        document.getElementById("quantity").value  
  
    );  
  
    const price = Number(  
  
        document.getElementById("price").value  
  
    );  
  
    const totalPriceElement =  
  
        document.getElementById("totalPrice");  
  
    const errorMessage =  
  
        document.getElementById("errorMessage");  
  
  
    if (quantity < 0 || price < 0 ||  
  
        Number.isNaN(quantity) || Number.isNaN(price)) {  
  
        errorMessage.textContent =  
  
            "Quantity and price must be valid non-negative values."  
  
        totalPriceElement.textContent = "₹0.00";  
  
    }  
}
```

7. CODE EXPLANATION

Code	Explanation
function calculateTotal()	Defines the function used to calculate the total price.
getElementById()	Accesses HTML elements using their IDs.
Number()	Converts input values into numbers.
if (...)	Checks whether quantity and price are valid non-negative values.
quantity * price	Calculates the total price.
textContent	Updates the total price displayed on the webpage.
addEventListener("input", ...)	Runs the calculation when quantity or price changes.
calculateTotal()	Performs the initial calculation when the page loads.

8. TEST CASES AND EXECUTION DATA

Test Case	Product	Quantity	Price	Expected Result	Actual Result	Status
TC-01	Pen	10	₹20	Total = ₹200.00	Total = ₹200.00	Pass
TC-02	Notebook	5	₹50	Total = ₹250.00	Not provided	—
TC-03	Pencil	0	₹10	Total = ₹0.00	Not provided	—

Test Case	Product	Quantity	Price	Expected Result	Actual Result	Status
TC-04	Book	2	₹125.50	Total = ₹251.00	Not provided	—

Additional Validation Test

Test Case	Input	Expected Result	Actual Result	Status
TC-05	Negative quantity	Validation message should be displayed.	Not provided	—
TC-06	Negative price	Validation message should be displayed.	Not provided	—

9. OUTPUT

Output 1: Product Calculator with Entered Values

Product Calculator

Product Name:

pen

Quantity:

10

Price per Unit (₹):

20

Total Price:

₹200.00

Output 2: Product Calculator Initial State

Product Calculator

Product Name:

Quantity:

Price per Unit (₹):

Total Price:

₹0.00

10. RESULT

Thus, the Interactive Product Calculator was implemented using HTML, CSS, and JavaScript. The supplied output screenshot verifies that a quantity of 10 and a unit price of ₹20 produce a total price of ₹200.00 without refreshing the webpage.

How this would be submitted

Source Code

Q1_Interactive_Product_Calculator/

- ├── index.html
- ├── style.css
- ├── script.js
- └── README.md